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with + or - sign. A real constant without any sign will be considered as positive.

Exponent form of real constants consists of two parts: **(1) Mantissa** and **(2) Exponent.** The mantissa must be either an integer or a real constant. The mantissa followed by a letter E or e and the exponent. The exponent should also be an integer.

For example, 58000000.00 may be written as 0.58×10^8 or 0.58E8.

Mantissa (Before E)	Exponent (After E)
0.58	8

Example:

5.864 E-1 ➔ 5.864×10^1 ➔ 58.64

5864 E-2 ➔ 5864×10^{-2} ➔ 58.64

0.5864 E-2 ➔ 0.5864×10^2 ➔ 58.64

Boolean Literals

Boolean literals are used to represent one of the Boolean values(True or false). Internally true has value 1 and false has value 0.

Character constant

A character constant is any valid single character enclosed within single quotes. A character constant in C++ must contain one character and must be enclosed within a single quote.

Valid character constants : 'A', '2', '$'

Invalid character constants : "A"

The value of a single character constant has an equivalent ASCII value. For example, the value of 'A' is 65.

Escape sequences (or) Non-graphic characters

C++ allows certain non-printable characters represented as character constants. Non-printable characters are also called as non-graphical characters. Non-printable characters are those characters that cannot be typed directly from a keyboard during the execution of a program in C++, for example: backspace, tabs etc. These non-printable characters can be represented by using escape sequences. An escape sequence is represented by a backslash followed by one or two characters.

Table 9.2 Escape Sequences

Escape sequence	Non-graphical character
\a	Audible or alert bell
\b	Backspace
\f	Form feed
\n	Newline or linefeed
\r	Carriage return
\t	Horizontal tab
\v	Vertical tab
****	Backslash
\'	Single quote
\"	Double quote
\?	Question Mark
\On	Octal number
\xHn	Hexadecimal number
\0	Null

Even though an escape sequence contains two characters, they should be enclosed within single quotes because, C++ consider escape sequences as character constants and allocates one byte in ASCII representation.